Chapter 7 Notes

Section 1

In the Remarks are page 91, here is the "easy induction": Obviously this holds for R_1. Now suppose that it holds for some R_n where $n > 1$. Then R_n has 2^{n-1} elements. By definition, $R_{n+1} = \mathcal{P}(R_n)$, which has $2^{2^{n-1}}$ elements, which is equal to 2^n, therefore completing the induction.

The proof of Proposition 1.3 is problematic. It uses Cantor's Theorem, though this hasn't even been stated (let alone proven) in the context of Zermelo-Frankel Universes. Proposition 1.3 can also be proven without using Cantor's Theorem. In particular, we first show that $R_\alpha \notin R_\alpha$.

Proof. By transfinite induction. Clearly this holds for 0. Suppose it holds for some α. Then, if $R_{\alpha+1} \in R_{\alpha+1}$, by definition of $R_{\alpha+1}$, $R_{\alpha+1}$ would be a subset of R_α. But $R_\alpha \in R_{\alpha+1}$. Thus this would mean that $R_\alpha \in R_\alpha$, contradicting the induction assumption. Hence $R_{\alpha+1} \notin R_{\alpha+1}$. On the other hand, suppose it held for all ordinals less than some limit ordinal λ. Then, by definition, $R_\lambda = \cup_{\alpha<\lambda} R_\alpha$. Suppose $R_\lambda \in R_\lambda$. Then, for some $\alpha < \lambda$, $R_\lambda \in R_\alpha$. But $R_{\alpha+1} \subset R_\lambda$, and $R_\alpha \in R_{\alpha+1}$, meaning $R_\alpha \in R_\lambda$. Thus, because R_α is transitive, $R_\alpha \in R_\alpha$, contradicting the induction assumption. Thus $R_\lambda \notin R_\lambda$, completing the transfinite induction. □

Then, in the proof of Proposition 1.3, $R_\alpha \neq R_{\alpha+1}$ follows for the following reason: We know that $R_\alpha \in R_{\alpha+1}$ and $R_\alpha \notin R_\alpha$ (by the above result), meaning $R_\alpha \neq R_{\alpha+1}$.

Section 2

Note that, at a minimum, the R_α sequence discussed in this section is an On-sequence of subsets of V (and of course, all On-sequences for any particular class are also On-sequences for V itself).

Note that the proof of O_6 uses Theorem 1.18 of Chapter 5.